NFC Attendance System – Final Project Report

Report of 17th February:

1. Introduction:

The NFC Attendance System is a Flutter-based desktop application designed to record attendance using NFC technology. The system integrates with a Google Sheets backend through a Google Apps Script web API.

2. System Architecture:

- **Components**
  - **Frontend:** Flutter (Windows desktop)
  - **Backend:** Google Apps Script (Web API)
  - **Database:** Google Sheets
- **Data Flow**
  1. NFC scan (or simulation) occurs in Flutter app
  2. Data is sent via HTTP POST request
  3. Google Apps Script receives the request (doPost)
  4. Data is stored in Google Sheets

3. Implementation Details

3.1 Flutter Application

- Built using Flutter framework
- Uses http package for API communication
- Main files:
  - main.dart → App entry point
  - nfc_scan_page.dart → NFC scan logic & UI

3.2 Google Apps Script Backend

- Handles HTTP POST requests
- Parses JSON data
- Writes to Google Sheet

Example structure:

- Entry Time
- Exit Time
- Student ID
- Room
- Type (IN/OUT)
- Method (SIM/TAG/HCE)

4. Development Progress :

🔹 **Phase 1: Initial Setup**

- Flutter project created
- Basic UI implemented
- Google Sheet prepared with columns

🔹 **Phase 2: Backend Integration**

- Apps Script created with doPost(e)
- Web app deployed
- URL generated

🔹 **Phase 3: Early Errors Encountered**

❌ **Error 1: Running URL in Terminal**

- URL pasted in PowerShell
- Error:

command not found

- Cause:
    - Terminal treats URL as command
- Fix:

- Use browser or:

start https://script.google.com/macros/s/.../exec

❌ Error 2: Script function not found: doGet

- Occurred when opening API URL in browser

Cause:

- Only doPost(e) was implemented
- Browser sends GET request

Fix:

- Add doGet() OR ignore (since app uses POST)

❌ Error 3: HTTP 302 (Moved Temporarily)

Displayed in Flutter app:

Error: 302
Moved Temporarily

Cause:

- Google Apps Script redirects request
- Flutter HTTP client did not follow redirect correctly
- Or wrong/old deployment URL used

❌ Error 4: No Data in Google Sheets

- API seemed working but no rows added

Possible Causes:

- Wrong Sheet name
- Incorrect Spreadsheet ID
- Deployment permissions not set to "Anyone"

5. Fixes Applied

✅ **Backend Fixes**

- Correct Spreadsheet ID and Sheet name
- Ensured doPost(e) correctly parses JSON
- Deployment updated:
  - Execute as: **Me**
  - Access: **Anyone**

✅ **API Fixes**

- Used correct **Web App URL**
- Ensured POST request (not GET)
- Used proper JSON body

✅ **Flutter Fixes**

- HTTP request corrected
- Error handling improved

6. Final Status (Latest Screenshots)

🟢 **Flutter Build & Execution**

- Command executed:

flutter run -d windows

- Output:

Built build\windows\x64\runner\Debug\nfc_attendance_system.exe

- Application launched successfully

🟢 **App Running**

- NFC Attendance UI visible
- No build errors
- Dart VM service running

🟢 **Main Application Code**

```
void main() {
 runApp(const MyApp());
}
```

- Entry point working correctly
- MyApp loads the main UI

7. Why Earlier Errors Disappeared

🔍 Root Causes vs Fixes

Problem	Root Cause	Why It Disappeared
302 Error	Redirect from Apps Script	Correct deployment URL + proper POST handling
doGet error	Browser sent GET request	App uses POST (doPost)
No data in sheet	Config mismatch	Fixed Spreadsheet ID + Sheet name
Terminal error	URL treated as command	Stopped using terminal for URL
API failure	Wrong permissions	Set access to **Anyone**

💡 Key Insight

Earlier errors were **not Flutter issues**, but:

- API misconfiguration
- Deployment mistakes
- HTTP method mismatch

Once backend and URL were fixed → everything worked.

8. Final System Status

Component	Status
Flutter App	✅ Running
Build System	✅ Stable
API Connection	✅ Working
Google Sheets	✅ Ready
NFC Simulation	✅ Functional

9. Conclusion

The NFC Attendance System is now:

- Fully functional
- Successfully integrated with Google Sheets
- Running smoothly on Windows

All previously encountered errors were resolved through:

- Correct API deployment
- Proper HTTP request handling
- Accurate configuration of backend services

10. Future Improvements

- Real NFC hardware integration
- Authentication system
- Dashboard for attendance tracking
- Error logging system